

Themes and Projects Overview

HCI Fundamentals

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Academic Year 2026/2027

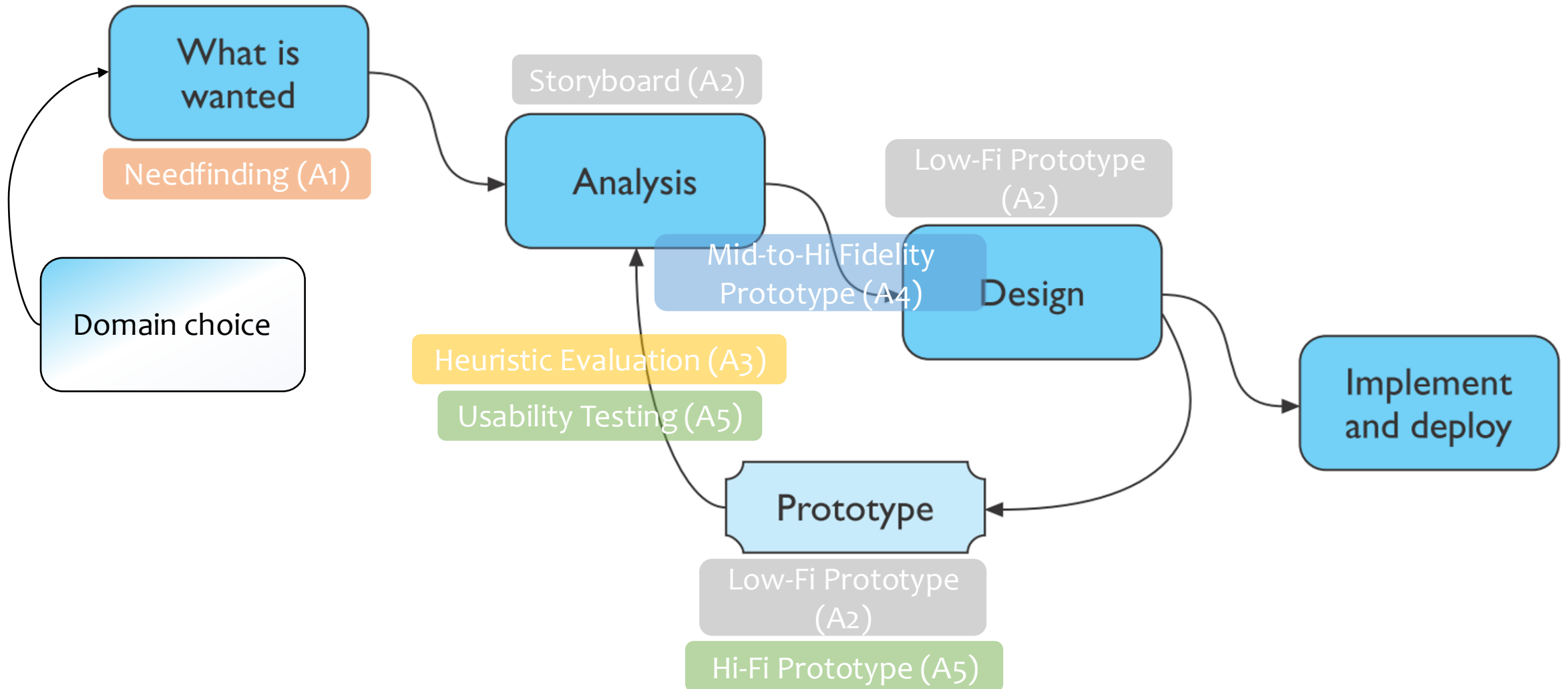
Project Development – Recap

- **Goal:** to give hands-on experience with the modern human-centered design process described during the course
- Projects will be built **step-by-step** and mostly carried on *during* labs
- Project's topic proposed by each group
 - Within the chosen theme and based on *needfinding*
- *Group assignments* represent the various process steps
 - Start during a lab
 - Work and discuss with teachers (in the following labs)
 - Evaluated at the exam through reports and discussion

(Planned) Assignments – Recap

- Assignment 1 [group]
 - *Needfinding*
- Assignment 2 [group]
 - *Storyboard and Low-fidelity prototype*
- Assignment 3 [individual]
 - *Heuristic evaluation on another group's low-fidelity prototype*
 - *Results passed to the other group*
- Assignment 4 [group]
 - *Medium-to-high fidelity prototype*
- Assignment 5 [group]
 - *High-fidelity prototype and usability evaluation*

Human-Centered Process meets *Assignments*



Projects Completion Level (I)

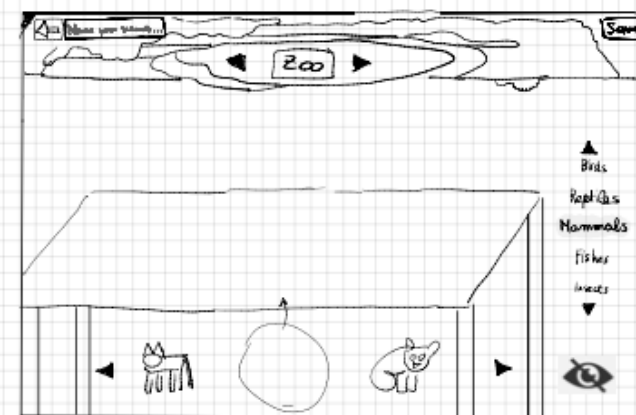
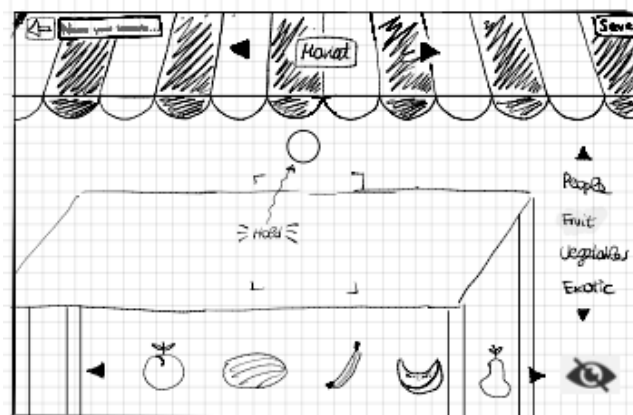
- The realized final prototype *must* be a **high-fidelity interactive prototype**
 - in code, with any technology you like/know
 - not a final “product”
 - simulating a realistic experience on a specific device
- For example, you can use web technology to built a prototype that *looks like* and *behave as* a mobile application

Projects Completion Level (II)

- The prototype will make some assumptions and will not implement standard or very complex features (even if technically important)
 - Which will be *faked*, *limited* in number, or *hard coded*
 - Your users will be already registered, signed in, and ready to “work” (no sign up, sign in, ... mechanism)
 - Any particular behavior will work in a small number of pre-defined cases
 - ...

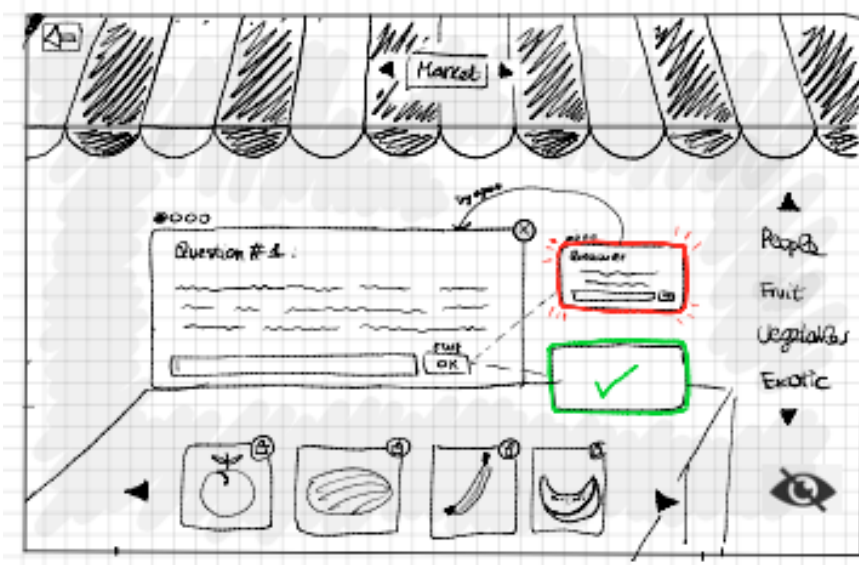
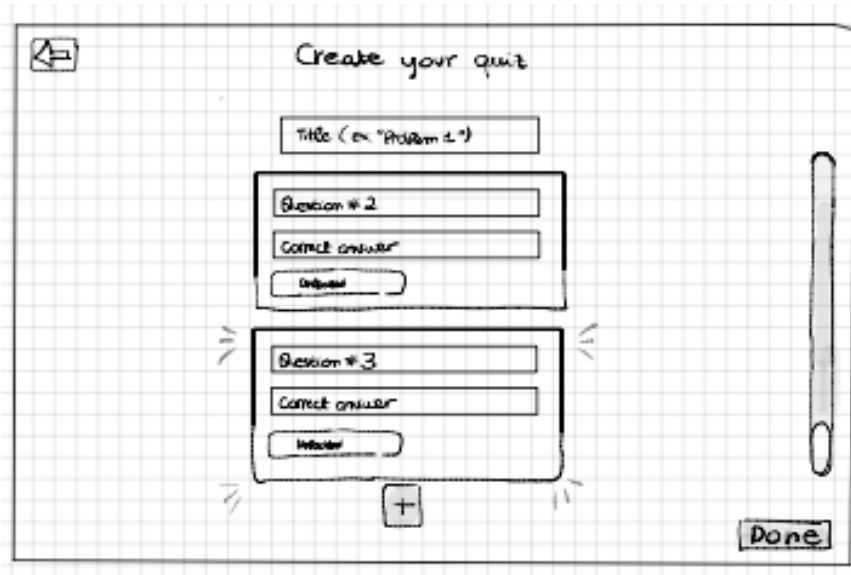
Example (2022) – Theme: ‘AR/VR for Education’

- Chosen domain: supporting elementary school teachers teaching math
- Picked solution (at the end of Assignment 1): “Allow teachers to create more engaging and better explaining scenarios to represent the [math] problem and the logic behind.”
- Excerpt from the 1st low-fi prototype:



Example (2022) – Theme: ‘AR/VR for Education’

- Excerpt from the 2nd low-fi prototype:



Example (2022) – Theme: ‘AR/VR for Education’

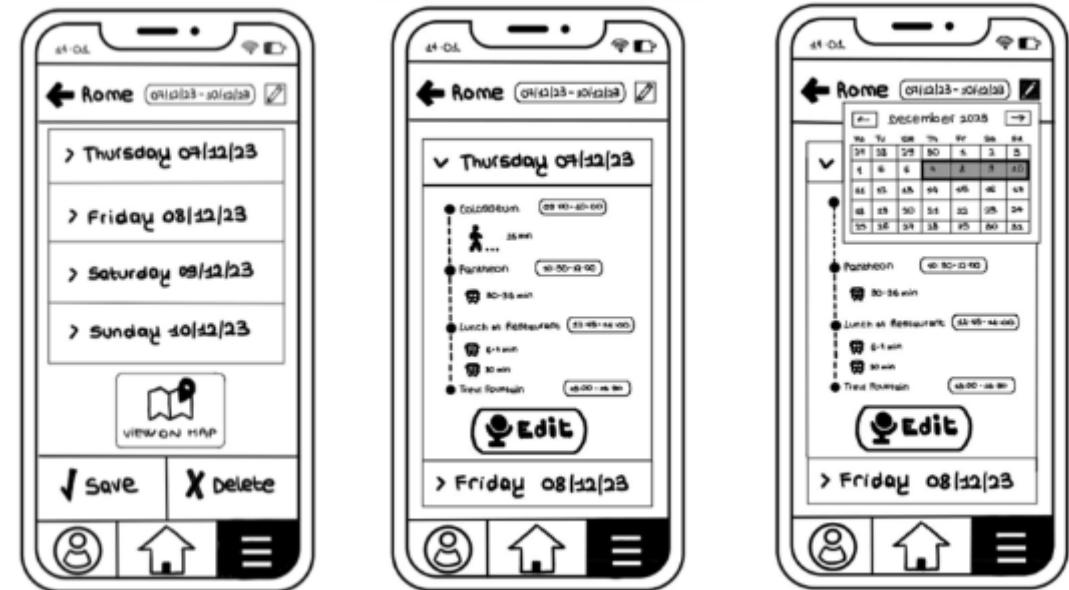
- Excerpts from the hi-fi prototype:



- Main limitations: no pitch-to-zoom, objects are put at the center of the scene, objects don't respond to the law of physics, objects are hard-coded in a JS file.

Example (2023) – Theme: ‘Humans meet AI’

- Chosen domain: urban tourism (national and international)
- Picked solution (at the end of Assignment 1): “Survey to understand user preferences and create the itinerary accordingly”
- Excerpt from the 1st low-fi prototype:



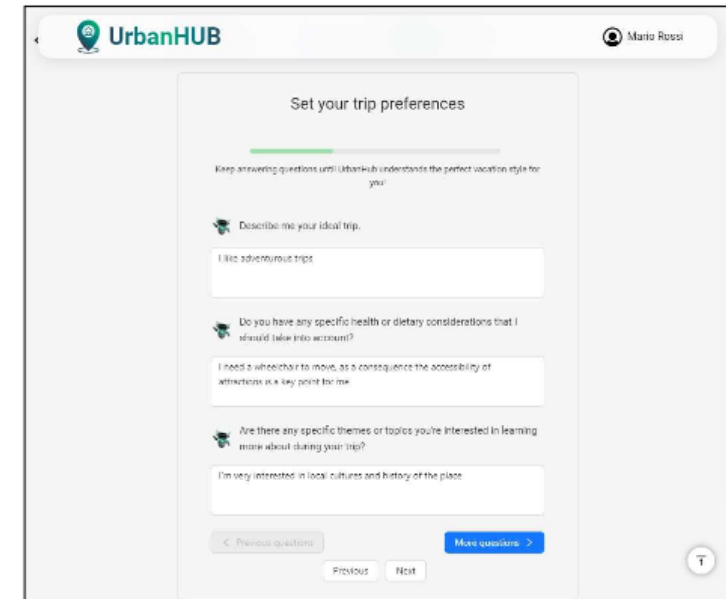
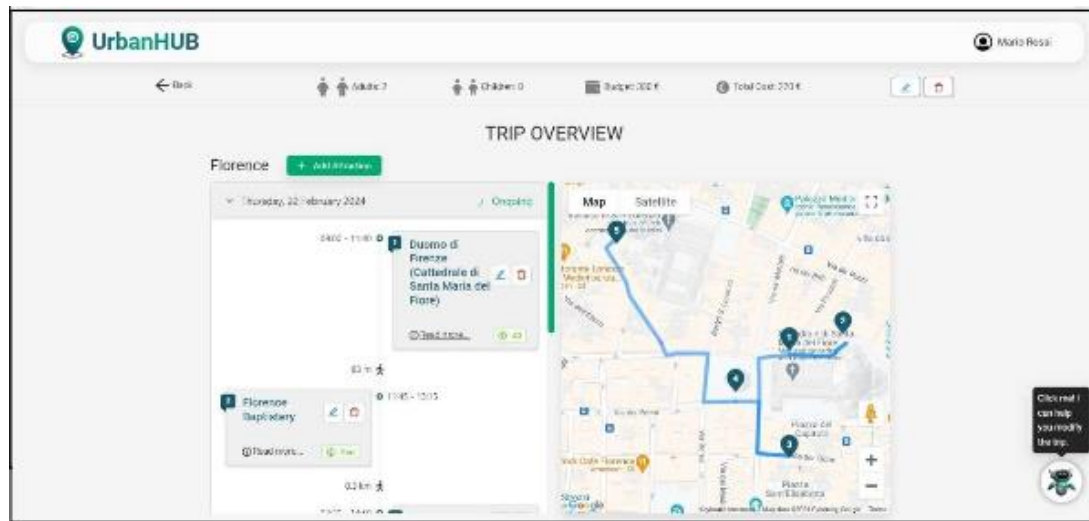
Example (2023) – Theme: ‘Humans meet AI’

- Excerpt from the 2nd low-fi prototype:



Example (2023) – Theme: ‘Humans meet AI’

- Excerpts from the hi-fi prototype:



- Main limitation: absence of a sophisticated AI driving the trip generation and reliance on a chatbot for edits. The chatbot is limited, as it only accepts a few specific input sentences for trip modifications, discarding other inputs.

This Year's Themes

Theme 1 – Playful Health and Wellbeing

- **Description:** Health and wellbeing are fundamental aspects of our daily existence. They encompass physical, mental, and emotional dimensions that can be significantly influenced by our interactions with technology. Within this theme, we will explore playful and innovative ways to create applications and interfaces that empower individuals to lead healthier lives.
- **When:** Friday 14:30-16:00 (A-L and M-Z)

Theme 1: Perspectives

- In thinking at the theme, you can ask yourselves:
 - Which health or wellbeing related topics we want to explore?
 - For whom?
 - Are we able to reach out to these people?
- Playfulness (see later) needs to be included from Assignment 2, not before
 - Right now, just keep it in the back of your mind
- Creativity and originality are *criteria* for the evaluation!
 - Helping students attending a local gym on their own is less original/creative than supporting them in using their smartphones mindfully and effectively managing their digital habits

Theme 2 – Playful Education and Learning

- **Description:** Education, either formal or informal, plays a pivotal role in many aspects of our life: you can learn at school, to play sports, to support class' learning activities, within families and to pass cultural traditions and languages. Within this theme, we will explore how we might create educational experiences for helping people learn better and possibly with more fun.
- **When:** Friday 16:00-17:30 (A-L), 17:30-19:00 (M-Z)

Theme 2: Perspectives

- In thinking at the theme, you can ask yourselves:
 - Which kind of education and learning do we want to focus on? At which scale (class, school, individuals, ...)? For formal or non-formal education?
 - For whom?
 - Are we able to reach out to these people?
- Playfulness (see later) needs to be included from Assignment 2, not before
 - Right now, just keep it in the back of your mind
- Creativity and originality are *criteria* for the evaluation!
 - Helping university students studying a difficult topic is less original/creative than supporting tourists to playfully learn the history of various monuments in the city

Theme 3 – Playful Exploration of the World

- **Description:** We are constantly “exploring the world”: we visit new places, we move to a new city or country to study, we get around our neighbors, we discover new monuments or museums, etc. Within this theme, we will focus on better supporting people exploring a bit of their world in any scale and environment (building, city, park, country, ...) by building playful applications for specific users.
- **When:** Friday 17:30-19:00 (A-L), 16:00-17:30 (M-Z)

Theme 3: Perspectives

- In thinking at the theme, you can ask yourselves:
 - Which aspects of “exploring the world” do we want to focus on? At which scale?
 - For whom? For which activities?
 - Are we able to reach out to these people?
- Playfulness (see later) needs to be included from Assignment 2, not before
 - Right now, just keep it in the back of your mind
- Creativity and originality are *criteria* for the evaluation!
 - Helping students exploring Politecnico alone is less original/creative than supporting visitors to discover typical cuisines of a city

Games and Gamification (in brief)

- Three central elements to play and games:
 1. **Dynamics.** General elements that define the patterns of how the game and the players will evolve over some time. The most popular game dynamics are narrative (good and engaging storyline), progression, and relationship.
 2. **Mechanics.** Action that moves the game forward, things that “makes the game an exciting and engaging one” like challenges, discoveries, collecting items, and rewards.
 3. **Game components.** Specific features that represent the intended mechanics and dynamics, like points, badges, and leaderboard.
- *Gamification*: the application of certain game mechanics to provide extrinsic motivation by offering intangible rewards, without other dynamics, to create user engagement.

What About Playfulness?

- Playfulness: an attitude of mind
 - according to John Dewey, a philosopher
- Gamification and games are *manifestations* of playfulness, but they not always create a playful mindset or experience on their own
- Playfulness is more about *dynamics*, the spontaneous, creative experience, and less on mechanics (which can be still present)
 - E.g., it can be found in humor, roaming around in a virtual environment or periodic reflection challenges
 - It increase effectiveness, motivation, and creativity
- In the themes, we want our users to manifest this mindset as much as possible

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